

Manufacturing Performance Dashboard – Power BI Project

Project type: End-to-end analytical dashboard (synthetic data)

Tools: Power BI, DAX, Python (synthetic data generation), Figma

Industry context: Automotive electronics manufacturing

GitHub Repository: www.github.com/laskowskaaleksandraa/manufacturing-powerbi-dashboard

1. Project Objective

Design and deliver a dashboard for monitoring and analysing manufacturing performance using a KPI framework based on OEE and its components.

2. Business Questions Addressed

- Where are we now compared to targets?
- Which production lines drive current losses?
- Are observed issues isolated events or emerging trends?
- Where should improvement actions be prioritised?
- How do local factors (teams, machines) contribute to line-level results?

3. Analytical Scope

- Availability losses: long downtime and Microstops
- Yield losses: scrap and First Pass Yield (FPY)
- Performance: included as a derived KPI, without dedicated loss analysis

Multi-level analysis:

- Factory level (overview and trends)
- Line level (root-cause exploration)
- Team and Machine level (local benchmarking)

4. Key Design Principles

- Non-additive KPIs handled with correct aggregation logic
- Operational-first design with strategic context available through KPI tooltips
- Local effectiveness and impact metrics designed to preserve line-level context
- Consistent interaction logic and tooltip design across dashboard pages
- Synthetic data generated with controlled variability to avoid artificial patterns

5. Assumptions & Validation

- Synthetic dataset based on realistic manufacturing structure
- Explicit documentation of simplifications and trade-offs
- KPI definitions aligned with standard OEE methodology

6. Deliverables

- Interactive Power BI dashboard
- Technical and analytical documentation
- Documented data model assumptions and KPI logic